

BIKE RENTAL DEMAND & USAGE ANALYSIS

Data Analytics Project

Tools: Python | SQL | Tableau

1. Executive Summary

I looked at bike rental data in order to understand the key factors that influence bike rental demand and to identify opportunities for better operational planning.

The dataset holds 10,886 observations of bike rental activity, which include details on time, season, weather, temperature, humidity, working days and customer type.

The analysis showed that bike rental demand varies a lot by time of day, season, temperature and weather conditions. Bike rental demand peaks during the evening period whereas bike rental demand is very low, in the morning. Fall records the highest average seasonal demand and clear weather is linked to the average bike rental activity.

Registered users make up 81.2% of total bike rentals making them the dominant customer segment. Casual-user participation drops significantly when weather is poor.

I think these findings give us opportunities to improve bike availability, maintenance scheduling, bike rental demand planning and customer segmentation.

2. Project Objectives

The project was designed to answer the following business questions:

- When does bike rental demand reach its peak and its lowest point?
- Which season generates the bike rental demand?
- How does weather affect bike rental activity?
- How does temperature relate to bike rental demand?
- What proportion of bike rentals come from casual users and from registered users?
- How does customer behavior change under different weather conditions, for bike rentals?
- How can these findings help make better operational decisions?

The overall objective was to move from bike rental data to actionable business insights and recommendations.

3. Data & Methodology

3.1 Data Used

The project uses a historical **Bike_Rental_1** containing **10,886 hourly records**.

The main variables used in the analysis include:

Data	Variables
Time	datetime, date, hour, month, day
Demand	count
Customer Type	Casual , Registered
Season	season
Weather	weather
Environment	temp, humidity, windspeed
Calendar	Working , Holiday

The **count** field represents total bike rentals, while **casual** and **registered** represent rentals from the two customer groups.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count					
2	1/1/2011 0:00	1	0	0	1	9.84	14.395	81	0	3	13	16					
3	1/1/2011 1:00	1	0	0	1	9.02	13.635	80	0	8	32	40					
4	1/1/2011 2:00	1	0	0	1	9.02	13.635	80	0	5	27	32					
5	1/1/2011 3:00	1	0	0	1	9.84	14.395	75	0	3	10	13					
6	1/1/2011 4:00	1	0	0	1	9.84	14.395	75	0	0	1	1					
7	1/1/2011 5:00	1	0	0	2	9.84	12.88	75	6.0032	0	1	1					
8	1/1/2011 6:00	1	0	0	1	9.02	13.635	80	0	2	0	2					
9	1/1/2011 7:00	1	0	0	1	8.2	12.88	86	0	1	2	3					
10	1/1/2011 8:00	1	0	0	1	9.84	14.395	75	0	1	7	8					
11	1/1/2011 9:00	1	0	0	1	13.12	17.425	76	0	8	6	14					
12	1/1/2011 10:00	1	0	0	1	15.58	19.695	76	16.9979	12	24	36					
13	1/1/2011 11:00	1	0	0	1	14.76	16.665	81	19.0012	26	30	56					
14	1/1/2011 12:00	1	0	0	1	17.22	21.21	77	19.0012	29	55	84					
15	1/1/2011 13:00	1	0	0	2	18.86	22.725	72	19.9995	47	47	94					
16	1/1/2011 14:00	1	0	0	2	18.86	22.725	72	19.0012	35	71	106					
17	1/1/2011 15:00	1	0	0	2	18.04	21.97	77	19.9995	40	70	110					
18	1/1/2011 16:00	1	0	0	2	17.22	21.21	82	19.9995	41	52	93					
19	1/1/2011 17:00	1	0	0	2	18.04	21.97	82	19.0012	15	52	67					
20	1/1/2011 18:00	1	0	0	3	17.22	21.21	88	16.9979	9	26	35					
21	1/1/2011 19:00	1	0	0	3	17.22	21.21	88	16.9979	6	31	37					
22	1/1/2011 20:00	1	0	0	2	16.4	20.455	87	16.9979	11	25	36					
23	1/1/2011 21:00	1	0	0	2	16.4	20.455	87	12.998	3	31	34					
24	1/1/2011 22:00	1	0	0	2	16.4	20.455	94	15.0013	11	17	28					
25	1/1/2011 23:00	1	0	0	2	18.86	22.725	88	19.9995	15	24	39					
26	1/2/2011 0:00	1	0	0	2	18.86	22.725	88	19.9995	4	13	17					
27	1/2/2011 1:00	1	0	0	2	18.04	21.97	94	16.9979	1	16	17					

Fig1. Raw Bike Rental Dataset

3.2 Data Preparation

The raw data was changed into a dataset that's ready for analysis by using Python.

The main changes were:

- Getting the hour the day and the month from the date and time field
- Changing the season codes into easy to understand season names
- Changing the weather codes into easy to understand weather names
- Making categories for working days and non-working days
- Finding the ratios of users and registered users
- Making groups, for temperature and humidity levels

1]EDA

```
(3) df.info()

(4) df['datetime'] = pd.to_datetime(df['datetime'])
    # changing datetime type to datetime from object
    df.info()

(5) df.describe()

(6) plt.boxplot(df['count'])
    plt.title('Outliers in the Rentals')
    plt.show
    #Here we can see there are outliers means there is very high demands in those periods, we won't remove outliers cause we need to make business decisions and those outliers are business spikes.
```

2]DATA EXPLORATION

```
# here we are gonna do feature engineering that gives insights

df['date'] = df['datetime'].dt.date
df['hour'] = df['datetime'].dt.hour
df['month'] = df['datetime'].dt.month
df['day'] = df['datetime'].dt.day_name()

df.head()
```

There are few columns right now in numbers but they actually represent category, so have to convert them in python/

```
# renaming the categorical columns

season_map = {1: 'Spring', 2: 'Summer', 3: 'Fall', 4: 'Winter'}
weather_map = {1: 'Clear', 2: 'Cloudy', 3: 'Light rain/snow', 4: 'Severe'}

df['season_label'] = df['season'].map(season_map)
df['weather_label'] = df['weather'].map(weather_map)

df.head()

df.isnull().sum() # this makes sure there are no null values in data
```

3]Feature Engineering

```
df['casual_ratio'] = df['casual'] / df['count']
df['registered_ratio'] = df['registered'] / df['count']

df['temp_bucket'] = pd.cut(df['temp'], bins=5)
df['humidity_bucket'] = pd.cut(df['humidity'], bins=5)

df.head()
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count	date	hour	month	day	season_label	weather_label	casual_ratio	registered_ratio	temp_bucket	humidity_bucket
0	2011-01-01 00:00:00	1	0	0	1	9.840000	14.385000	81	0.000000	3	13	10	2011-01-01	0	1	Saturday	Spring	Clear	0.187500	0.812500	(8.858, 10.892]	(80.0, 100.0]
1	2011-01-01 01:00:00	1	0	0	1	9.020000	13.835000	80	0.000000	8	32	40	2011-01-01	1	1	Saturday	Spring	Clear	0.200000	0.800000	(8.858, 10.892]	(60.0, 80.0]
2	2011-01-01 02:00:00	1	0	0	1	9.020000	13.835000	80	0.000000	5	27	32	2011-01-01	2	1	Saturday	Spring	Clear	0.156250	0.843750	(8.858, 10.892]	(60.0, 80.0]
3	2011-01-01 03:00:00	1	0	0	1	9.840000	14.385000	75	0.000000	3	10	13	2011-01-01	3	1	Saturday	Spring	Clear	0.230769	0.769231	(8.858, 10.892]	(60.0, 80.0]
4	2011-01-01 04:00:00	1	0	0	1	9.840000	14.385000	75	0.000000	0	1	1	2011-01-01	4	1	Saturday	Spring	Clear	0.000000	1.000000	(8.858, 10.892]	(60.0, 80.0]

4] Answering Business Questions and visualizations

Q1) What factors affecting the Demands of Rentals?

Figure 2. Data Transformation and Feature Engineering Using Python

3.3 Data Validation

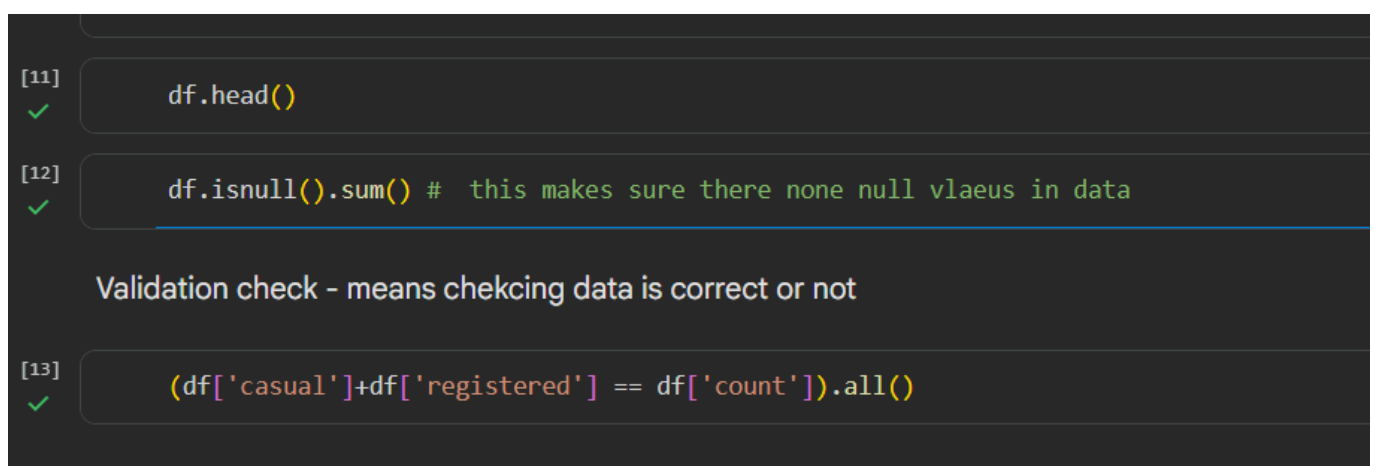
Data-quality checks were performed before analysis, including:

- Missing-value checks
- Duplicate checks
- Data-type checks
- Logical consistency checks

A key validation was:

Casual Rentals + Registered Rentals = Total Rentals

The dataset passed these checks, allowing the analysis to proceed using the prepared data.



```
[11] ✓ df.head()

[12] ✓ df.isnull().sum() # this makes sure there none null vlaeus in data

Validation check - means chekcing data is correct or not

[13] ✓ (df['casual']+df['registered'] == df['count']).all()
```

Figure 3. Data Quality Validation

3.4 Analytical Tools

- I used Python to prepare data build features, explore data and find correlations.
- I used SQL to answer business questions by summarizing data grouping records applying conditions and comparing customers.
- I used Tableau to build the interactive dashboard and to present the main findings with visuals.
- Overall workflow:

Raw Data → Python → SQL → Tableau → Business Insights

4. Key Findings & Visualizations

Finding 1 — Rental Demand Peaks During Specific Hours

In this graph below the x axis shows the hour of the day from 0 to 23. The y axis shows the average rentals for the books.

We can see two peak periods :-

- Morning Peak: 7 to 9 AM.
- Evening Peak: 4 to 7 PM.

Business meaning:

1. Capacity Planning

The business should make sure that the maximum number of bikes are available when customers need them most. That means during the times of 7–9 AM and 4–7 PM. In those windows the demand is higher so having enough bikes will keep customers satisfied.

2. Maintenance Planning

Instead of treating the whole low-demand period as the only time, for maintenance the business can choose any shorter low-demand window between:

- 9PM-5AM

These windows let maintenance teams work when very few people are renting bikes. At the time the main morning and evening peaks are avoided, so the service does not interfere with high-demand periods.

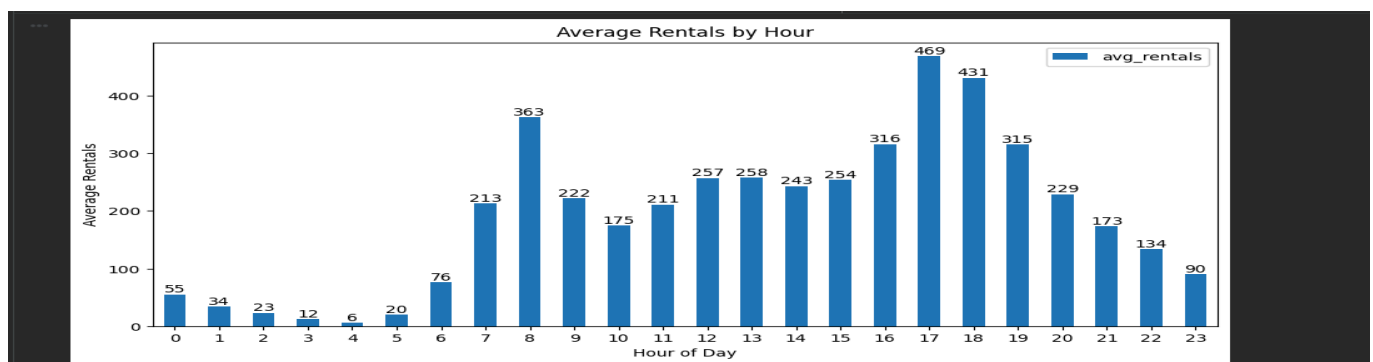


Figure 4. Average Bike Rental Demand by Hour

Finding 2 — Rental Demand Varies by Season

The graph shows that Fall has the average rental demand, followed by Summer, Winter and Spring. The graph also shows that the customer mix changes across seasons indicating usage patterns throughout the year.

Business Meaning —

- Capacity Planning: Make sure there are bikes and staff during busy seasons, especially Fall and Summer so that no rental chances are lost.
- Seasonal Promotions: Offer discounts, special deals or seasonal packages during lower-demand times such, as Spring to encourage more rentals and use the bikes more. Registered Customers range is almost same throughout all seasons, focus on casual customers more.

In this graph below the x axis shows the Seasons of the Year from . The y axis shows the average rentals for the booking..

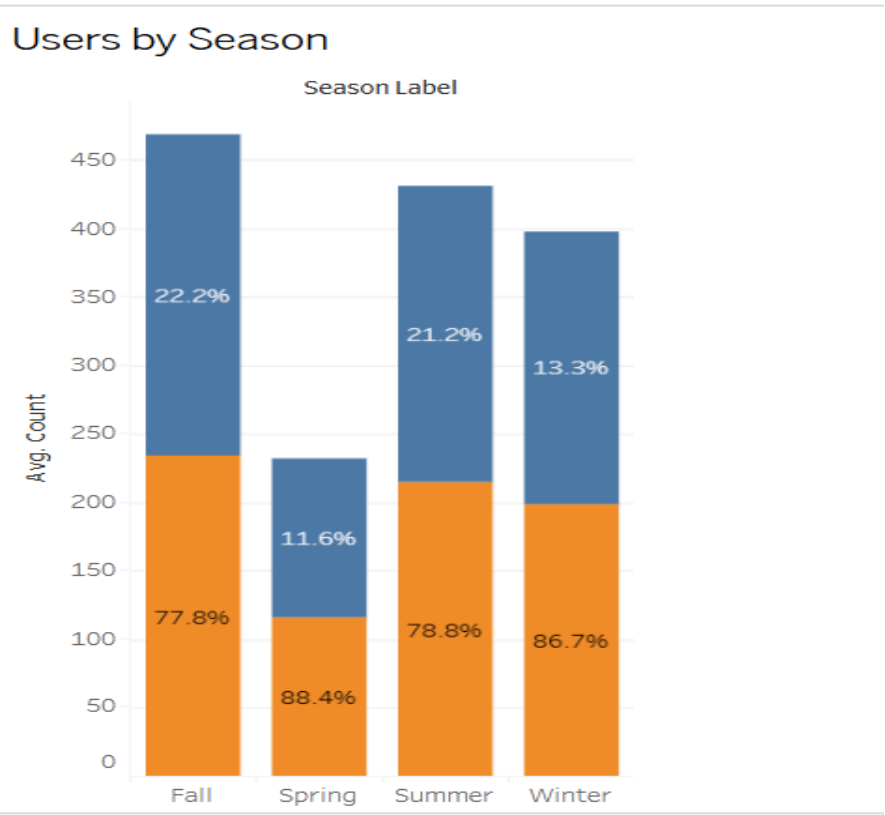


Figure 5. Average Bike Rental Demand by Season

Finding 3 — Temperature Is Positively Associated With Demand

Temperature is linked to rental demand in a positive way with a correlation of about +0.394. Average demand rises as temperature rises hitting a peak in the 30 to 40 degrees Celsius range, where around 334 rentals are seen.

Business Meaning :-

- Temperature can be added to demand planning with other factors, like time, season and weather.
- Adjust Operating Hours: When the temperature is good the business could open hours or offer more service to meet extra demand.
- Targeted Marketing: Use temperature based campaigns for example by advertising rides when the weather feels comfortable.

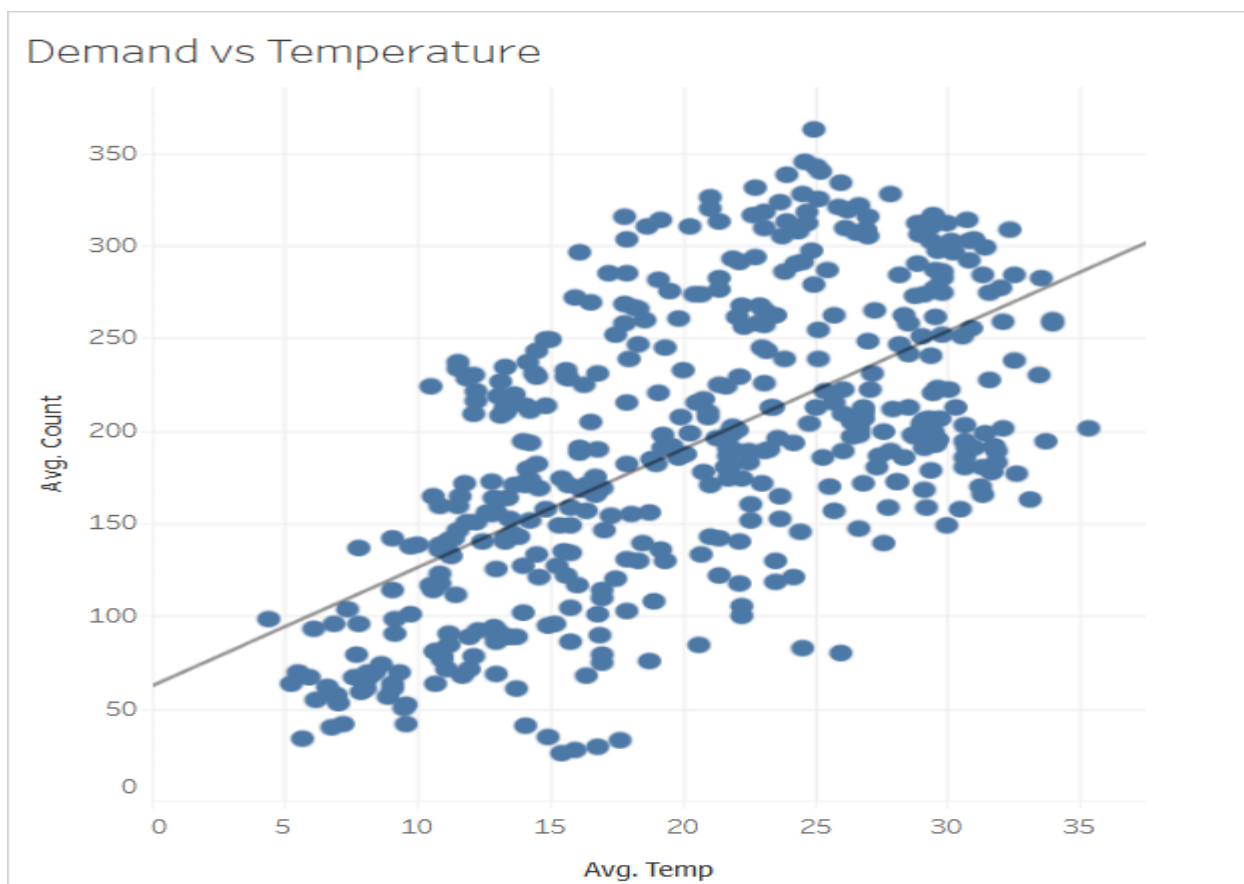


Figure 6. Relationship Between Temperature and Rental Demand

Finding 4 — Weather Strongly Affects Casual User Participation .

We can see that the registered-user share remains relatively high across all weather conditions increasing from 82 percent in weather to 96 percent in severe weather.

In contrast the casual-user share steadily decreases as weather becomes worse:

Clear → 18%

Cloudy → 16%

Light Rain/Snow → 13%

Severe → 4%

This shows that casual users are more sensitive to unfavorable weather conditions while registered users continue to form the majority of rentals.

Business meaning :

- Casual Customer Targeting: Focus marketing efforts on users during favorable weather when casual users are more likely to rent bikes.
- Weather-Based Promotions: Introduce targeted offers or weekend packages for users when weather conditions are favorable to encourage additional rentals.
- Registered Customer Retention: Since registered users remain the core customer base across weather conditions maintain engagement and retention programs, for this stable segment.

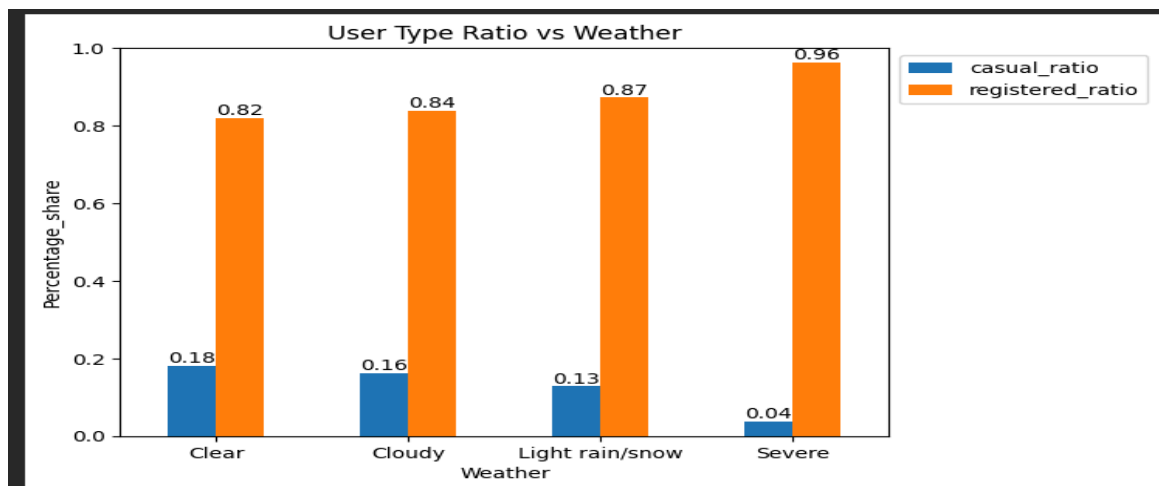


Figure 6. User Type Ratio Across Weather Conditions

Finding 5 — Registered Users Dominate Overall Rentals

Registered users make up 81.2 percent of all rentals while casual users make up about 18.8 percent. This shows that registered customers are the part of the customer base and provide most of the rental activity.

Business meaning :-

- Customer Retention: Pay attention to keeping registered users by offering membership benefits and loyalty programs.
- Membership Conversion: Reach out to casual users, with membership trials or introductory plans to turn them into registered customers.
- Customer Segmentation: Use the behaviors of casual users and registered users to develop more precise marketing strategies.

```
63
64
65 -- Q5 Causal v/s Register user by percentages
66
67 • SELECT
68     sum(casual) as Total_Causal,
69     sum(registered) as Total_Registered,
70     ROUND(sum(casual)*100/sum(count), 1) AS Causal_Percentage,
71     ROUND(sum(registered)*100/sum(count), 1) AS Registered_Percentage
72 FROM bike_rental_1 ;
73
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	Total_Causal	Total_Registered	Causal_Percentage	Registered_Percentage
▶	392135	1693341	18.8	81.2

Result 9 x

Output

Figure 8. SQL Analysis of Casual vs Registered User Contribution

5. Tableau Dashboard

The final Tableau dashboard brings together the KPIs and visualizations from the analysis into one interactive business view.

The dashboard provides a view of:

- Overall Performance: Total Rentals, Average Rentals per Hour
- Peak Demand: Peak Rental Demand and Peak Demand Hour
- Seasonality: Best Season
- Demand Patterns: Demand by Hour and Weekday/Hour Heatmap
- Environmental Factors: Demand vs Temperature and Average Demand by Weather
- Customer Analysis: Users by Weather and Total Rentals, by User Type

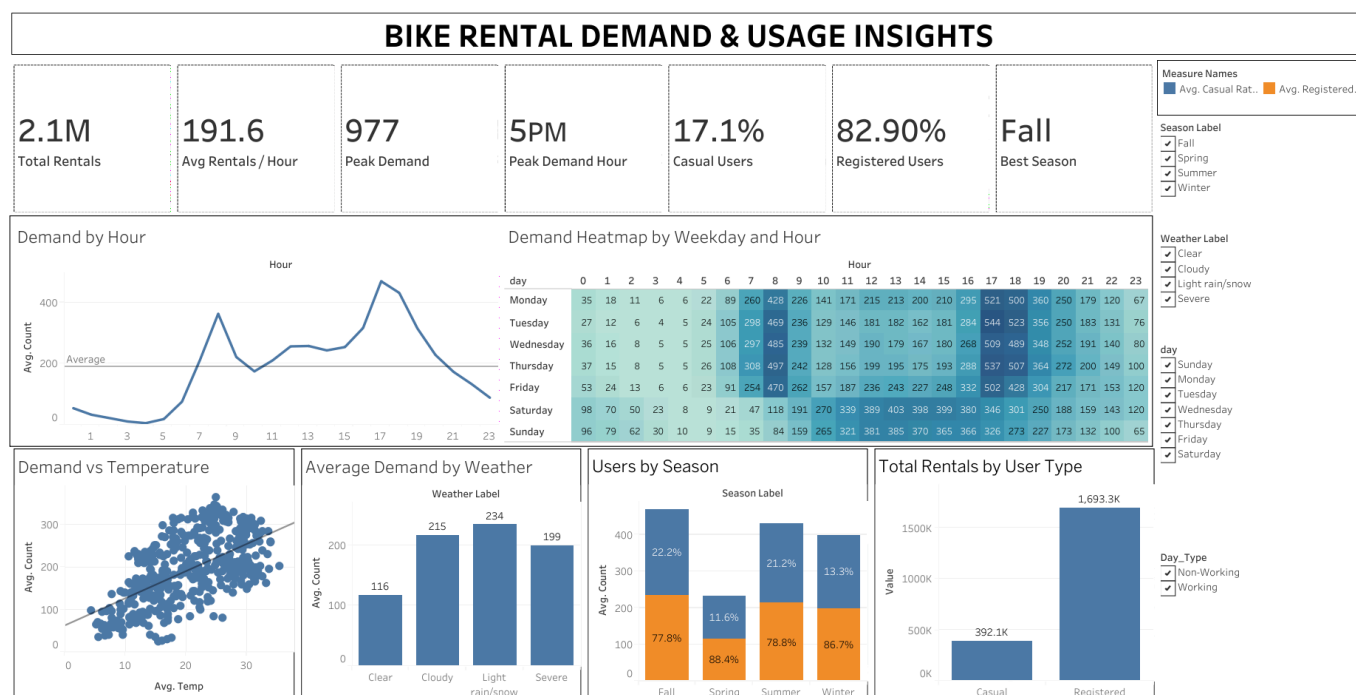


Figure 9. Final Tableau Dashboard — Bike Rental Demand & Usage Insights

6. Actionable Recommendations

1. Capacity & Maintenance Planning

Capacity & Maintenance Planning must keep bikes ready during the two main demand peaks: 7–9 AM and 4–7 PM. Capacity & Maintenance Planning helps customers rent when they want and cuts down on missed rental chances.

For maintenance Capacity & Maintenance Planning should schedule work periods during low demand between 9 PM and 5 AM. Capacity & Maintenance Planning keeps service work out of the morning and evening peaks.

2. Seasonal Capacity & Promotions

Seasonal Capacity & Promotions must bring bikes and staff during high-demand seasons, especially Fall and Summer to seize more rental chances.

During lower-demand times like Spring, Seasonal Capacity & Promotions can add discounts, special deals or seasonal packages to push rentals and better bike use.

Registered customers stay about the all seasons so Seasonal Capacity & Promotions can lean more on casual customers for extra business.

3. Temperature-Based Planning & Operations

Temperature-Based Planning & Operations should mix temperature with time, season and weather when estimating demand.

When temperatures are good Temperature-Based Planning & Operations can lengthen hours. Boost service to catch more rentals.

Temperature-Based Planning & Operations can also run campaigns to advertise bike rides under pleasant conditions.

4. Weather-Based Casual Customer Strategy

Weather-Based Casual Customer Strategy should target customers when weather is good because they are more ready to rent bikes.

Weather-Based Casual Customer Strategy can launch weather-based offers, weekend packages or focused promotions to pull casual rentals.

At the time registered customers stay the main base in all weather so Weather-Based Casual Customer Strategy must keep retention and engagement programs for this steady group.

5. Customer Retention & Membership Conversion

Customer Retention & Membership Conversion relies on registered users who bring rentals so Customer Retention & Membership Conversion must keep them happy, with membership perks and loyalty schemes.

Customer Retention & Membership Conversion should also reach users with membership trials, starter plans or other conversion offers to turn them into registered customers.

Customer behavior can guide Customer Retention & Membership Conversion to split groups accurately and deliver precise marketing.

7. Conclusion

The analysis shows that bike rental demand is strongly influenced by time of day, season, temperature, weather and customer type.

The hourly analysis identified morning and evening peak periods creating opportunities for better capacity planning while low-demand hours can be used for maintenance activities. Seasonal analysis showed demand during Fall and Summer while lower-demand periods such as Spring provide opportunities for seasonal promotions.

Temperature and weather patterns can be incorporated into demand planning and operating decisions. The weather analysis also showed that casual customers are more sensitive to weather while registered customers remain the core and more stable customer segment. This creates opportunities, for targeted promotions, customer retention and converting users into registered members.

Overall the project converts bike rental data into practical business insights and actionable recommendations covering capacity planning, maintenance planning, seasonal and weather-based marketing, operating decisions, customer retention and membership conversion.